

Statement of Differences Between the Conference Version and the Extended Version

Mona Rajhans

Vishal Khawarey

The conference version of this paper [?] presented an empirical study of adversarial robustness and SHAP-based explainability drift for a Multi-Layer Perceptron (MLP) classifier evaluated on two cybersecurity datasets: a phishing URL detection dataset and the UNSW-NB15 network intrusion dataset. Adversarial robustness was measured under the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD), and the Robustness Index (RI) was introduced as a scalar summary of the accuracy–perturbation curve. The paper further characterised the coupling between gradient-based feature sensitivity and SHAP attribution drift, and demonstrated that adversarial training partially recovers robustness on both datasets.

This extended version makes the following substantive contributions that together constitute well over 30% new material relative to the conference paper.

(i) New classifier families. Random Forest and XGBoost are added alongside the MLP, representing the tree-ensemble family that dominates production cybersecurity deployments but was absent from the conference study. Both models are evaluated under identical experimental conditions across all datasets and attack methods.

(ii) New attack methods. Three black-box attacks are introduced and evaluated: ZOO [?] (finite-difference gradient approximation), Square Attack [?] (score-based random search), and HopSkipJump [?] (decision-boundary estimation). This extends the attack surface from gradient-based white-box attacks, which are inapplicable to non-differentiable tree models, to a fully model-agnostic black-box evaluation framework.

(iii) New metric: Explainability Stability Index (ESI). ESI is defined as the complement of the normalised area under the SHAP-drift curve, providing an explanation-domain analogue of RI that operates on the same $[0, 1]$ scale. ESI is computed via TreeSHAP [?] for Random Forest and XGBoost, enabling direct cross-model comparison of prediction robustness and explanation stability within a single unified framework.

(iv) New datasets. The NF-ToN-IoT dataset [?] (1.16 M NetFlow v2 records, 10 features) is added as a third evaluation domain covering IoT network intrusion detection. HIKARI-2021 [?] (555,278 flows, 83 features) is added as a fourth domain covering encrypted TLS/HTTPS traffic detection, providing additional cross-domain validation of all findings across structurally different feature spaces.

(v) New empirical findings. Evaluation reveals that ZOO produces near-degenerate gra-

dient estimates against XGBoost ($RI \approx 0.98$) due to its piecewise-constant prediction surface, whereas Square Attack reduces XGBoost accuracy to near-random ($RI = 0.36$ on Phishing). This 0.62 gap exposes a methodological risk: attack-method selection is not model-agnostic. Additionally, we demonstrate that XGBoost achieves high prediction robustness under ZOO ($RI \approx 0.98$) yet exhibits low explanation stability ($ESI \approx 0.06$ – 0.16 across all four datasets), showing that prediction robustness and explanation stability are *decoupled*—a high RI does not guarantee trustworthy explanations under attack. To confirm the RF $\hat{}$ XGB ESI ordering is not itself a ZOO artefact, we recompute ESI under Square Attack across all four datasets. The ordering holds on Phishing (RF = 0.230, XGB = 0.117) and UNSW-NB15 (RF = 0.184, XGB = 0.155), is tied on NF-ToN-IoT, and reverses on HIKARI-2021 due to a normalisation-scale artefact (XGBoost’s absolute SHAP drift is $\approx 10\times$ larger than RF’s, but grows more gradually, which the normalised-AUC formula scores as higher stability). The architectural interpretation is confirmed on three of four datasets.

(vi) PGD step-size ablation. A controlled ablation over PGD step sizes $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.10\}$ confirms that the PGD < FGSM anomaly observed in the conference paper is a hyperparameter sensitivity artifact on z-score normalised tabular data, not a structural property of the model. PGD with $\alpha = 0.01$ (the conventional choice) yields $RI = 0.82$ versus FGSM’s $RI = 0.68$ on Phishing; the gap closes only at $\alpha \geq 0.05$. This provides a rigorous empirical response to reviewer feedback on the conference version.

(vii) Master comparative analysis. A comprehensive results table spanning all three datasets, three model families, and five attack methods is introduced, enabling cross-domain and cross-model comparison in a single view. Eleven new figures accompany the extended experiments: the ZOO-degeneracy accuracy curve, full attack comparison curves (ZOO, Square Attack, HopSkipJump), the master RI/ESI heatmap, per-feature TreeSHAP drift heatmaps for both tree models on Phishing, NF-ToN-IoT robustness curves, HIKARI-2021 robustness curves, the PGD step-size ablation, ESI drift curves comparing ZOO versus Square Attack on Phishing, ESI drift curves on UNSW-NB15 confirming the RF $\hat{}$ XGB ordering under Square Attack, and the Square Attack adversarial training before/after curves for tree ensembles.

(viii) Feature-level TreeSHAP drift analysis. A new subsection decomposes the aggregate ESI score into per-feature attribution drift, revealing which feature types drive explanation instability for each model family. The analysis confirms that XGBoost’s boosted-additive structure produces higher per-feature drift than Random Forest’s ensemble averaging, consistent with the consistently lower XGBoost ESI observed across all three datasets.

(ix) Attack-method selection guidance. A new discussion section introduces a two-axis framework (gradient dependence, query efficiency) that explains the observed attack ranking across all three datasets and yields concrete practical recommendations: Square Attack should be the primary benchmark for tree ensembles; ZOO alone overestimates robustness by up to 0.62 RI; adaptive PGD step sizes ($\alpha = \varepsilon/steps$) are recommended for z-score normalised tabular data.

(x) Variance and significance study. A five-seed variance study (varying train/test split, model initialisation, and evaluation subsets) is added to the Limitations section. The ZOO degeneracy gap is highly reproducible (mean 0.534–0.575, $std \leq 0.035$ across all four datasets),

confirming the primary finding is not a lucky point estimate. ESI variance is higher (RF ESI std up to 0.101); the RF > XGB ordering holds in all five seeds on UNSW-NB15 and in 4/5 seeds on Phishing and HIKARI-2021, but is within noise on NF-ToN-IoT (0.025 ± 0.082).

(xi) Square Attack adversarial training for tree ensembles. As a black-box analogue of MLP FGSM adversarial training, Square Attack data augmentation (25% adversarial ratio at $\varepsilon_{\max} = 0.3$) is evaluated on Phishing and UNSW-NB15. The largest beneficiary is XGBoost on Phishing, where Square RI rises from 0.372 to 0.755 (+0.383), nearly closing the gap with baseline RF. RF on UNSW-NB15 improves by +0.156, preventing the complete accuracy collapse (0.000) observed above $\varepsilon = 0.167$ in the baseline. XGBoost on UNSW-NB15 shows cross-attack transfer (+0.045 ZOO, +0.081 HSJ), while RF on Phishing is slightly degraded (-0.049), revealing that augmentation at $\varepsilon_{\max}$ can backfire for already-robust models. Clean accuracy cost is at most 0.7% across all conditions.

(xii) Limitations section. A dedicated Limitations section discusses five scope constraints absent from the conference version: feature-validity constraints on ℓ_{∞} perturbations in structured security data, dataset scope, CPU-only computational budget, the approximation of ESI via ZOO-perturbed inputs, and the variance study results quantifying which conclusions are statistically robust.

All tables, figures, and textual descriptions in this extended version are either entirely new or substantially revised. Results from the conference paper are summarised in prose with explicit citation [?] rather than reproduced verbatim. The title, abstract, and conclusions have been rewritten to reflect the extended scope of the work. The Related Work section has been expanded to position ESI explicitly against existing explanation-stability metrics (OpenXAI RIS/ROS, Alvarez-Melis and Jaakkola sensitivity measures) and to address the gap in tree-ensemble robustness literature. Experiment code and result tables are publicly available at [?].